

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	B.tech AIML-2023	Date:	06-08-2026

Code of Conduct

I P. Lohith Krishna (192325122) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

QUESTIONS: 5

Total Marks: 50

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

Answer: A healthcare company wants to detect pneumonia from chest X-ray images. Initially, the company considers using Machine Learning (ML) with manually extracted image features. After collecting millions of X-ray images, the company switches to Deep Learning (DL) because it is better suited for large image datasets.

Machine Learning requires manual feature extraction, where experts identify important image features such as edges, textures, or shapes before training the model. The model's performance depends heavily on the quality of these manually selected features. Machine Learning generally works well with small or medium-sized datasets and requires less computational power.

In contrast, Deep Learning automatically learns important features directly from raw images using deep neural networks such as Convolutional Neural Networks (CNNs). It eliminates the need for manual feature engineering and can learn complex patterns that may not be visible to human experts.

Deep Learning requires a large amount of data, which is available in this scenario because the company has collected millions of medical images. It also requires powerful hardware such as GPUs or TPUs for faster training. Although the computational cost is higher, Deep Learning usually provides better accuracy in image classification tasks.

One of the major advantages of Deep Learning is its ability to improve continuously as more data becomes available. In medical image analysis, it can detect subtle patterns associated with pneumonia, leading to more accurate diagnoses. However, Deep Learning models require longer training time, higher computational resources, and are often more difficult to interpret than traditional Machine Learning models.

Overall, Deep Learning is the better choice for this application because it automatically extracts image features, performs well on very large datasets, achieves higher prediction accuracy, and is widely used in real-world medical imaging systems despite its higher computational requirements.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex realworld datasets.

Answer: Representation Learning is a technique in Deep Learning where the model automatically learns useful and meaningful features from raw data instead of relying on manually designed features. The learned features, also called representations, help the model understand patterns and relationships in the data.

In this scenario, the retail company has millions of customer purchase records and wants to identify customer buying patterns automatically. Instead of manually selecting features such as purchase frequency, average spending, or product preferences, a Deep Learning model can learn these important patterns directly from the raw transaction data.

In traditional feature engineering, data scientists must carefully analyze the dataset and manually choose features that they believe are useful for prediction. This process requires domain knowledge, takes significant time, and may overlook important hidden relationships in the data.

With Representation Learning, the model automatically discovers meaningful features during training. It can learn complex customer behaviors, purchasing trends, seasonal preferences, and hidden relationships among products without manual intervention. As more data becomes available, the learned representations become more accurate and informative.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

Answer:

A startup is developing a handwritten digit recognition system, and two different neural network architectures are proposed. One uses a **very wide network** with a single hidden layer containing thousands of neurons, while the other uses a **deep network** with multiple hidden layers and fewer neurons in each layer.

The **width** of a neural network refers to the number of neurons in each hidden layer, whereas the **depth** refers to the number of hidden layers. A wide network can learn many features at one level

and is simpler to train. However, it often requires a large number of neurons and parameters, increasing memory usage and the risk of overfitting.

A deep network learns features in multiple stages. The first hidden layers learn simple features such as edges and lines, while deeper layers learn more complex patterns like shapes and complete digits. This hierarchical learning makes deep networks more effective for image recognition tasks.

In terms of computational complexity, a very wide network requires more memory because of the large number of neurons. A deep network may take longer to train due to multiple layers, but it usually uses parameters more efficiently and achieves better feature learning.

Answer: Activation functions introduce non-linearity into neural networks, allowing them to learn complex patterns. Different activation functions are suitable for different layers of a deep learning model.

The Sigmoid activation function produces output values between 0 and 1. It is mainly used in the output layer of binary classification problems. However, it suffers from the vanishing gradient problem, making it less suitable for hidden layers.

The Tanh activation function produces outputs between -1 and 1. It converges faster than Sigmoid because its outputs are centered around zero. However, it can also suffer from vanishing gradients in deep networks.

The ReLU (Rectified Linear Unit) is the most commonly used activation function in hidden layers. It is computationally efficient, speeds up training, and reduces the vanishing gradient problem. However, some neurons may become permanently inactive, causing the dying ReLU problem.

Leaky ReLU is an improved version of ReLU. It allows a small output for negative inputs, preventing neurons from becoming inactive. It is a good choice when dead neurons are observed during training.

ELU (Exponential Linear Unit) also addresses the dying ReLU problem and often provides faster and more stable learning by allowing negative outputs.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question:

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

Answer: Unsupervised training is a learning approach in which a neural network learns patterns from unlabeled data without requiring target outputs. It is especially useful when collecting labels is expensive or impractical.

In this scenario, the social media platform collects billions of unlabeled user images every month. Instead of manually labeling every image, the neural network can automatically discover hidden patterns and meaningful representations from the available data.

During training, the model analyzes similarities and differences among images to learn useful features. It identifies common shapes, colors, textures, and visual patterns without knowing the actual image labels. These learned features can later be used for image classification, recommendation systems, or image retrieval.

Compared to supervised learning, unsupervised learning has several advantages in this scenario:
It does not require manually labeled data.

It saves both time and cost.

It can learn from very large datasets.

It discovers hidden patterns that may not be obvious to humans.

It creates useful feature representations for future supervised tasks.

Several techniques can be used for representation learning in unsupervised neural networks, including:
Autoencoders, which learn compressed representations of images.

Restricted Boltzmann Machines (RBMs), which learn hidden features from input data.

Self-Supervised Learning, where the model creates its own learning tasks from unlabeled data.

Generative models, which learn the underlying structure of the data.

6. Restricted Boltzmann Machines (RBM)

An online music streaming platform wants to improve its recommendation system by learning hidden relationships between users and songs from historical listening data without using labeled information.

Question:

Explain how a Restricted Boltzmann Machine (RBM) can be used in this recommendation system.

Describe the architecture of an RBM, its working principle, training process, and discuss its advantages and limitations in real-world recommendation applications.

7. Autoencoders

A manufacturing company installs hundreds of sensors on industrial machines to continuously monitor their health. Most of the collected sensor data represents normal machine behavior, while faulty conditions occur very rarely. The company wants an AI system that can automatically identify abnormal machine behavior without requiring labeled fault data.

Question:

Discuss how an Autoencoder can be used for anomaly detection in this scenario. Explain the architecture of an Autoencoder, the roles of the encoder, bottleneck layer, and decoder, and describe how reconstruction error helps detect abnormal conditions.

8. Deep Learning Applications

A smart city project aims to integrate multiple AI-based services, including face recognition for security, speech recognition for public information kiosks, medical image analysis in hospitals, autonomous traffic monitoring, and intelligent language translation for tourists.

Question:

Explain how Deep Learning can be applied in each of these services. Discuss the advantages of Deep Learning in solving these real-world problems and identify the challenges that may arise during deployment.

9. Selecting the Appropriate Deep Learning Technique

A technology company is planning to build an AI-powered platform with the following features:

- Image classification for uploaded photos
- Customer recommendation system
- Data compression for cloud storage
- Feature extraction from unlabeled data

The management asks the AI team to recommend suitable Deep Learning techniques for each task.

Question:

Identify the most appropriate Deep Learning models or techniques for each feature. Justify your recommendations by explaining the working principles and advantages of Representation Learning, Autoencoders, Restricted Boltzmann Machines, and other relevant Deep Learning concepts.

10. Designing a Deep Learning Solution

A national disaster management agency wants to build an intelligent emergency response system capable of analyzing satellite images, identifying damaged buildings, classifying disaster severity, and providing rescue recommendations. The system must process large amounts of image data accurately and efficiently.

Question:

Design a Deep Learning-based solution for this application. Explain whether Machine Learning or Deep Learning is more suitable, discuss the importance of Representation Learning, recommend appropriate activation functions for the network, and suggest whether unsupervised techniques such as Autoencoders or Restricted Boltzmann Machines could improve the overall performance of the system. Support your answer with appropriate reasoning and real-world examples.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandabl e but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____